

IT-Simplera Institute



Week 1 Task Report

Submitted By

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Domain

Network Administration

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Networking Fundamentals & Network Simulation Tools

Local Area Network

Introduction

During the first week of my Network Administration Internship at **IT-SIMPLERA Institute**, I learned the basic ideas about computer networking and got practical experience by using Cisco Packet Tracer and GNS3. The goal of this task was to figure out how devices talk to each other in a Local Area Network (LAN), set up IP addresses, and check if connections are working using the ping command.

Networking Concepts Learned

- **Computer Network:** A group of interconnected devices that communicate and share resources.
- **LAN (Local Area Network):** Connects devices within a limited geographical area such as an office or laboratory.
- **WAN (Wide Area Network):** Connects networks across cities or countries.
- **Switch:** Connects devices within a LAN and forwards frames using MAC addresses.
- **Router:** Connects different networks and forwards packets based on IP addresses.
- **IP Address:** A logical address assigned to each device for communication.
- **Subnet Mask:** Defines the network and host portions of an IP address.
- **MAC Address:** A unique hardware address assigned to every network interface.
- **Ping:** A network utility used to test connectivity between devices.

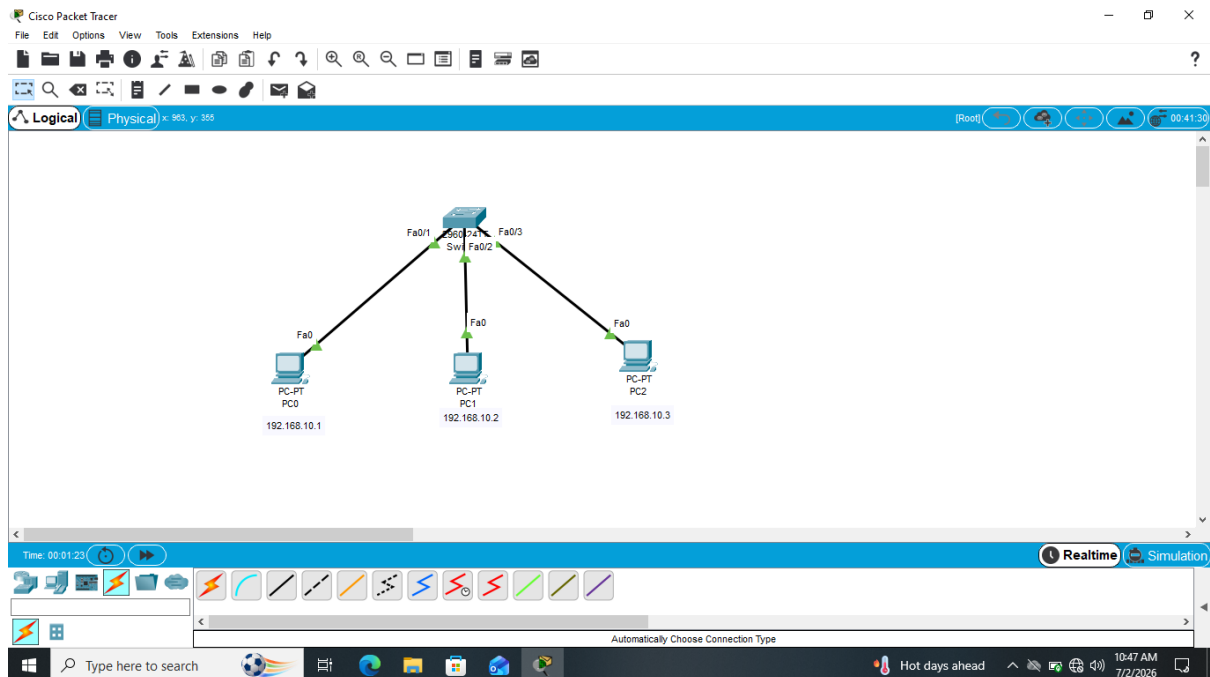
Communication Models: Networking follows standard models like OSI and TCP/IP. These models divide network communication into layers. Each layer has a specific role in sending and receiving data.

Ping Command: Ping is used to test connectivity between two devices. It sends small packets of data and checks if a response is received normally four packets are sent to test the echo.

Cisco Packet Tracer

Cisco Packet Tracer is a powerful network simulation tool developed by Cisco Systems, designed for learning, practicing, and visualizing computer networks, IoT, and cybersecurity concepts. It offers a virtual environment to create, configure, and troubleshoot network topologies with drag-and-drop simplicity, making it ideal for students, educators, and IT professionals. Also known as a network simulator, Packet Tracer supports multi-user collaboration and real-time protocol simulation.

LAN Network Topology :



Connectivity Testing

PC1: IP:192.168.10.1

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=39ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time=3ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 39ms, Average = 11ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=15ms TTL=128
```

PC2: IP: 192.168.10.2

PC1

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=2ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=3ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 1ms

C:\>ping 192.168.10.3

Pinging 192.168.10.3 with 32 bytes of data:

Reply from 192.168.10.3: bytes=32 time=32ms TTL=128
```

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PC3: IP: 192.168.10.3

PC2

Physical Config Desktop Programming Attributes

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time=1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time<1ms TTL=128
Reply from 192.168.10.1: bytes=32 time=45ms TTL=128

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 45ms, Average = 11ms

C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time=1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
```

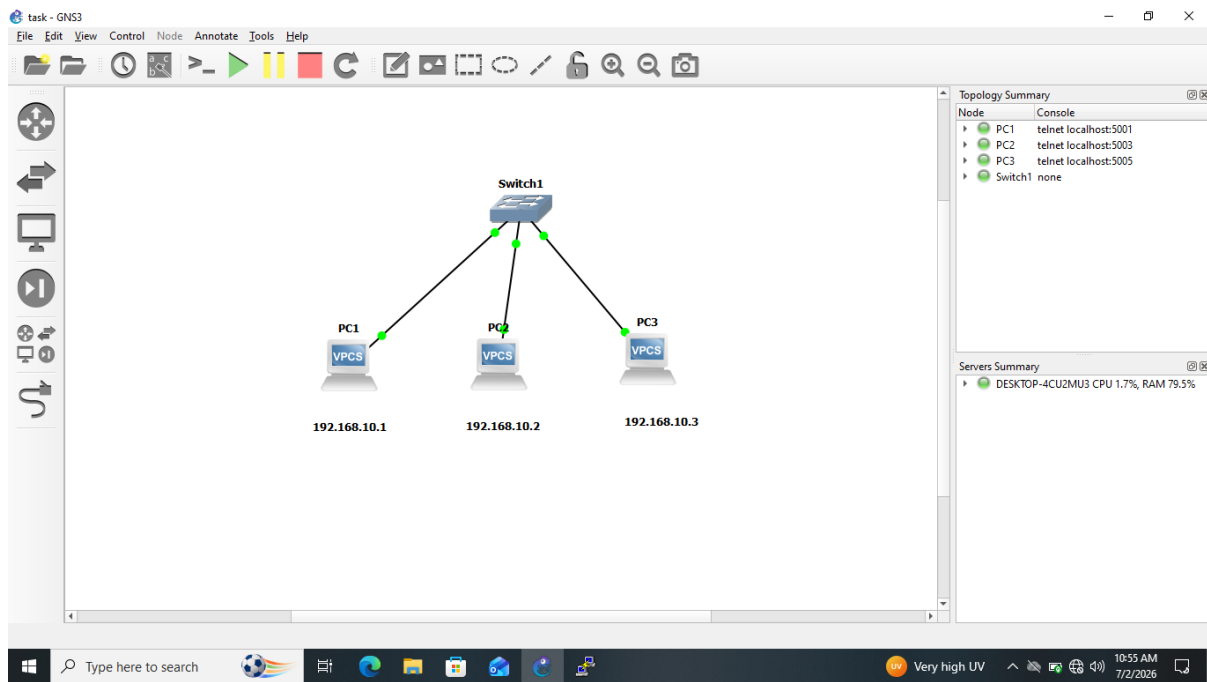
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GNS3

GNS3 is an advanced network emulation platform that allows engineers to build realistic network environments. I recreated the same LAN topology using one Ethernet switch and three VPCS nodes. Static IP addresses were assigned to all VPCS devices, and connectivity was verified using the ping command from the console.



Ping Test Results (Gns3)

PC1: IP: 192.168.10.1

task - GNS3

File Edit View Control Node Annotate Tools Help

PC1 - PuTTY

Dedicated to Daling.
Build time: Apr 10 2019 02:42:20
Copyright (c) 2007-2014, Paul Meng (mirnshi@gmail.com)
All rights reserved.

VPCS is free software, distributed under the terms of the "BSD" licence.
Source code and license can be found at vpcs.sf.net.
For more information, please visit wiki.freecode.com.cn.

Press '?' to get help.

Executing the startup file

PC1> ip 192.168.10.1/24
Checking for duplicate address...
PC1 : 192.168.10.1 255.255.255.0

PC1> ping 192.168.10.2
84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=1.265 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=1.427 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=1.748 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=1.766 ms

Topology Summary

Node	Console
PC1	telnet localhost:5001
PC2	telnet localhost:5003
PC3	telnet localhost:5005
Switch1	none

Servers Summary

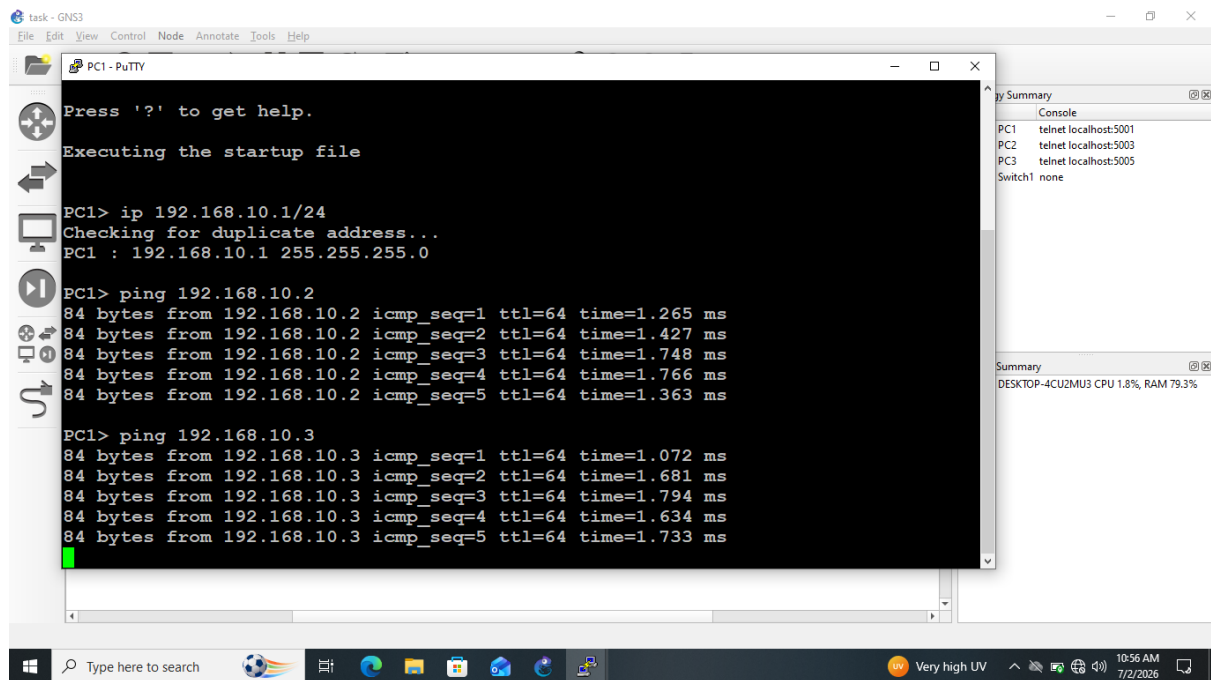
DESKTOP-4CU2MU3 CPU 2.0%, RAM 79.2%

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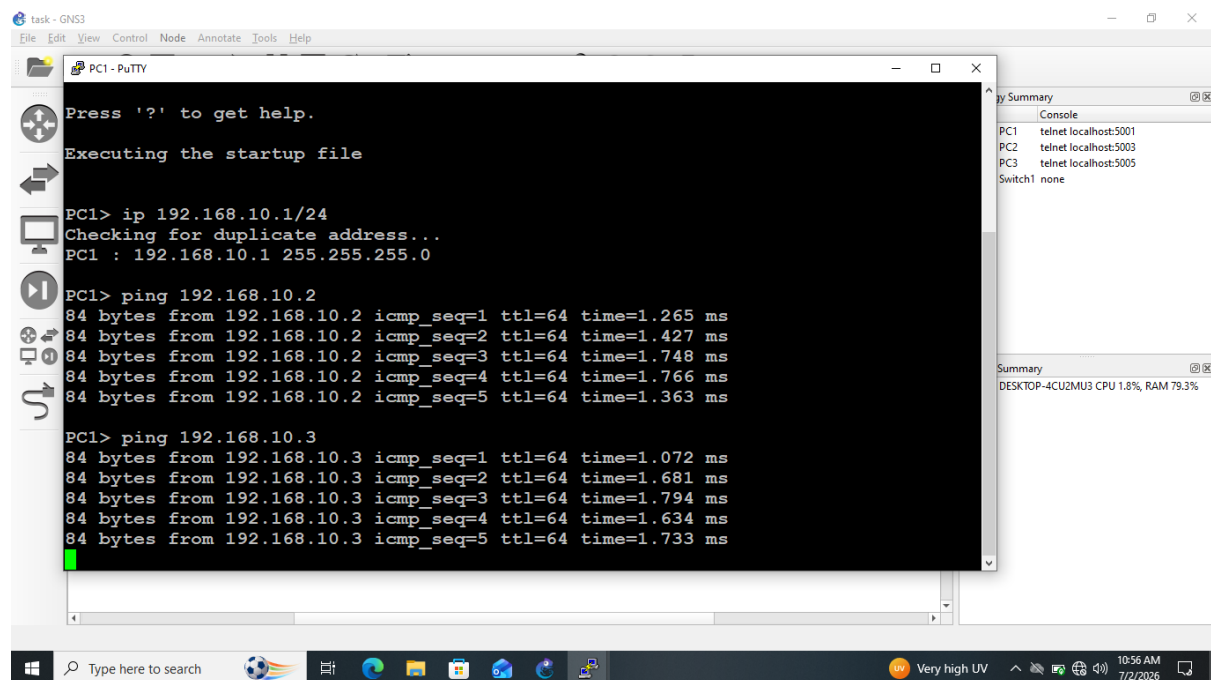
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PC2: IP: 192.168.10.2



PC3: IP: 192.168.10.3



Challenges Faced

Understanding the GNS3 interface.

- Configuring VPCS IP addresses.
- Verifying correct cable connections.
- Performing connectivity tests and confirming successful communication.

Conclusion

This Task strengthened my understanding of basic networking concepts and provided practical experience with two industry-standard simulation tools. I successfully designed a LAN topology, configured static IP addresses, and verified end-to-end connectivity using the ping utility. This practical exercise improved my confidence in network configuration, troubleshooting, and documentation.
